

PROJECT DEMONSTRATION PHASE

ComicCraft – AI Comic Story Creator using Gemini Models

1. Demonstration Overview

The Project Demonstration Phase presents the working flow of ComicCraft from user input to the generation and export of an AI-created comic. The demonstration shows the web interface, AI text generation, image generation, comic layout, preview, and PDF export.

2. Demonstration Objectives

The demonstration focuses on the following objectives:

- Show how a user provides a story prompt and comic preferences.
- Demonstrate structured story-outline generation using Gemini Flash.
- Demonstrate detailed narration and dialogue generation using Gemini Pro.
- Demonstrate AI-generated comic panel illustrations.
- Display the generated comic through the web interface.
- Demonstrate the final PDF export process.

3. Demonstration Environment

Item	Demonstration Detail
Application	ComicCraft – AI Comic Story Creator
Backend	FastAPI
Frontend	HTML, CSS and Jinja2
Text AI	Gemini Flash and Gemini Pro
Image AI	Stable Diffusion
Local server	Uvicorn at 127.0.0.1:8000

4. Demonstration Flow

The demonstration follows the documented processing sequence: user input → structured outline → story narration and dialogue → panel image generation → layout building → web presentation → PDF export.

PROJECT DEMONSTRATION PHASE – USER INPUT AND STORY GENERATION

1. Opening the ComicCraft Interface

The demonstration begins by starting the FastAPI application through Uvicorn and opening the local ComicCraft interface. The homepage provides the input form through which the user defines the comic story requirements.

2. Entering Comic Requirements

The demonstration shows the five documented input categories: Story Prompt, Main Character Name, Setting, Story Tone, and Art Style. These fields provide the information required to guide AI generation.

Input	Demonstration Purpose
Story Prompt	Defines the main story idea or scenario.
Main Character Name	Identifies the central character.
Setting	Defines where the story takes place.
Story Tone	Specifies the desired storytelling tone.
Art Style	Specifies the requested visual style.

3. Generating the Structured Outline

After submission, Gemini Flash is demonstrated as the first text-generation stage. Its documented responsibility is to create a structured five-panel outline, establishing the sequence of events that will become the comic panels.

4. Demonstration Checkpoint

At this stage, the presenter can verify:

- The submitted inputs are accepted by the backend.
- The outline generation stage is reached.
- The generated content follows the five-panel structure.
- The workflow proceeds to detailed story generation.

PROJECT DEMONSTRATION PHASE – NARRATION, DIALOGUE AND IMAGES

1. Detailed Story Generation

The second AI text-generation stage uses Gemini Pro to expand the structured outline into detailed narration and character dialogue. During the demonstration, the presenter can show how the initial outline becomes richer story content suitable for a comic.

2. Panel Illustration Generation

The image-generation stage uses Stable Diffusion through the project's image-generation component. The generated image corresponds to the panel content and requested art style. The images are stored in the project's static/panels location.

3. Demonstration Sequence

Step	Demonstration Action	Expected Output
1	Send the user inputs.	Comic generation workflow starts.
2	Generate outline.	Five-panel structured outline.
3	Expand story.	Narration and dialogue.
4	Generate images.	Panel illustrations.
5	Build layout.	Combined comic structure.

4. Image Generation Test

The /test-image route provides a documented demonstration point for testing image generation from a custom prompt. This route can be used separately from the complete comic workflow to confirm that the image-generation component is functioning.

5. Demonstration Checkpoint

The presenter can verify that:

- Story narration and dialogue are generated after the outline.
- Panel images are produced from the generation process.
- Generated images are available for the layout stage.

PROJECT DEMONSTRATION PHASE – COMIC PREVIEW AND EXPORT

1. Building the Comic Layout

After text and image generation, the layout-building stage combines the generated material into a comic presentation. The documented `build_comic_layout()` function is responsible for preparing the comic layout structure.

2. Web Comic Preview

The `comic_preview.html` template is used to present the generated comic content through the web interface. The demonstration can show the panel arrangement, generated illustrations, narration, and dialogue as the final comic is prepared for export.

3. PDF Export Demonstration

The final stage uses the project's `save_pdf()` function in `exporters.py` to create the comic PDF. Generated exports are documented under `static/exports`. The demonstration shows that the completed comic is converted from the generated layout into an exportable document.

4. Export Demonstration Checklist

- Confirm that the comic layout has been constructed.
- Open the comic preview page.
- Check that generated panels are visible.
- Check narration and dialogue associated with the panels.
- Trigger the PDF export process.
- Confirm that the export-success page is reached.

5. Main Routes Used During Demonstration

Route	Demonstration Role
/	Open the ComicCraft homepage and input form.
/generate	Run the main comic-generation workflow.
/generate-comic/json	Demonstrate structured JSON-based generation.
/export-success	Show successful export result.
/test-image	Independently test image generation.

6. Demonstration Result

The expected demonstration result is a complete AI-generated comic that can be viewed through the web interface and exported as a PDF.

PROJECT DEMONSTRATION PHASE – VALIDATION AND PRESENTATION SUMMARY

1. Demonstration Validation

The demonstration validates the major user-visible stages of ComicCraft without changing the documented project workflow. Each stage provides a checkpoint for confirming that the corresponding component has produced the expected output.

Stage	Validation Point
Input	All documented comic preferences are accepted.
Outline	A structured five-panel outline is produced.
Story	Narration and character dialogue are generated.
Images	Panel illustrations are generated.
Layout	Text and images are combined into the comic structure.
Preview	The comic is displayed through the web template.
Export	The comic is saved as a PDF and export-success page is shown.

2. Presenter Sequence

1. Introduce ComicCraft and its purpose.
2. Open the local application.
3. Enter the five comic-generation inputs.
4. Generate the structured outline and detailed story.
5. Show generated panel illustrations.
6. Display the comic preview.
7. Export the comic as a PDF.
8. Show the export-success result.

3. Technical Demonstration Points

The demonstration highlights the relationship between FastAPI routes, Jinja2 templates, Gemini text generation, Stable Diffusion image generation, layout construction, and PDF export. This provides a complete view of how the documented project components work together.

4. Demonstration Outcome

The Project Demonstration Phase presents ComicCraft as an end-to-end AI comic-generation system. The demonstration covers the documented path from user requirements to generated comic preview and PDF export, providing a practical presentation of the project's implemented workflow.

Conclusion

A successful demonstration shows the major features of ComicCraft in sequence and provides a clear practical representation of the project's architecture, AI integration, user interface, and final comic-generation output.